

Charotar University of Science and Technology
M.Sc. (Nano Science & Technology) Semester I Examination April 2010

NT701 Concepts of Physical Sciences I

Date: 3rd May 2010

Time: 10.00 am to 10.30 pm

Max. Marks 20

Part – A

Do as directed (Show your calculation within the space provided)

Que.1 In NaCl structure the distance between Na and Cl is 1nm, then the lattice constant (1)
(i) $\frac{1}{2}$ nm (ii) 2 nm (iii) 1 nm (iv) 4nm

Que.2 The atomic radius of fcc lattice is (1)
(i) $a/2$ (ii) $\sqrt{2} a/4$ (iii) $\sqrt{3} a/4$

Que.3 An fcc crystal has an atomic radius of 0.1246 nm, the d_{200} spacing is (Show the calculation and tick the correct answer) (3)

(i) $1.76\text{\AA}$ (ii) $1.24\text{\AA}$ (iii) $3.52\text{\AA}$ (iv) $2.03\text{\AA}$
Que.3 The momentum of a photon of wavelength $2\text{\AA}$ is: (show the calculation and tick the correct answer) (3)

(i) $2.3 \times 10^{-24} \text{ kg.m/s}$ (ii) $3.3 \times 10^{-24} \text{ kg.m/s}$ (iii) $4.3 \times 10^{-24} \text{ kg.m/s}$

Que.4 Write the Fermi-Dirac energy distribution Function. (2)

Que.5 The crystals with small energy gap will have the electron mobility (1)

- (i) Low (ii) High (iii) does not depend

Que.6 When metal transit from normal state to superconducting state the latent heat of material: (1)

- (i) Changes
(ii) Does not change
(iii) Changes but difficult to detect

Que.7 The Fermi energy of silver is 5.51 eV. What is the average energy of a free electron at 0K (show the calculation and tick the correct one) (2)

- (i) 2.026 eV (ii) 3.306 eV (iii) 1.096 eV

Que.8 The Fermi energy at 0K lies: (1)

- (i) Above the conduction band
(ii) Below the valence band
(iii) Between conduction and valence band.

Que.9 The lead material works as superconductor at a temperature $T_c = 7.26\text{K}$. If the constant characteristics of the lead material at 0 K is $H_0 = 8 \times 10^5 / 4\pi$, what is the value of the magnetic field in the lead at 5K (Show calculation and tick the correct one) (3)

- (i) $1.18 \times 10^4 \text{A/m}$ (ii) $2.18 \times 10^4 \text{A/m}$ (iii) $3.18 \times 10^4 \text{A/m}$

Que.10 A magnetic field of high strength is applied to Photon in the x-direction, what will be the effect on photon direction: (1)

- (i) Opposite to the applied field (ii) Perpendicular to applied field (iii) No change

Que.11 Ohm's law in vector form is: (1)

- (i) $J = \sigma E$ (ii) $J = \sigma / E$ (iii) $E = J\sigma$

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Part B

(Write each section on separate answer book)

Section I

- Que 1. (a) State Bragg's condition of XRD. (1)
(b) Draw the planes of a cubic crystal with proper axis having Miller indices of [111], [100], and [001]. (3)
(b) Find the angle between the two planes [100] and [1,1,0]. (3)
(c) The Bragg angle corresponding to the first order reflection from [311] planes in a fcc crystal is 35° when X-rays of wavelength $1.54 \text{ \AA}$ are used. Calculate the inter-atomic spacing. (3)
- Que 2. (a) Explain Langevin's theory of paramagnetism. (6)
(b) Why magnetic domains forms in ferromagnetic substances? (4)
- OR
- (a) Explain superconductivity behavior using BCS theory (6)
(b) Discuss the effect of magnetic field in a superconducting transition (4)

Section II

- Que 1. (a) What are Eigen values and Eigen functions? Write down the Schrödinger's equation for a particle in the box. (5)
(b) A particle is confined to one-dimensional infinite potential well of width 0.2 nm . It is found that when the energy of the particle is 230 eV its eigen functions have 5 antinodes. Find the mass of the particle. $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. (5)
- Que 2. (a) What is a grand canonical ensemble? (3)
(b) The density of Zinc is $7.13 \times 10^3 \text{ kg/m}^3$ and its atomic weight is 65.4, calculate the Fermi energy and the mean energy at $T = 0\text{K}$. (5)
(c) Define Fermi energy. (2)
- Que 3. (a) What are acceptor levels? (2)
(b) How will you determine the band gap of an intrinsic semiconductor? (5)
(c) Determine the number density of donor atoms which have to be added to an intrinsic semiconductor to produce an n-type semiconductor of conductivity $5 \text{ } \Omega^{-1} \text{ cm}^{-1}$. Given the mobility of electrons is $3850 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$. (3)
- OR
- Que 3. (a) How does the conductivity of an intrinsic semiconductor vary with temperature? (4)
(b) An intrinsic semiconductor material P has an energy gap of 0.36 eV , while another material Q has an energy gap of 0.72 eV . Compare the intrinsic carrier densities in these materials at 300K . Assume that the effective masses of the electron and holes are equal to the free electron mass. (6)
